

Lizzy Amaya

D1 – Design Ideas

Idea Description How It Meets Requirements (from C)

1. Elevated Basking Platform v2.0 (New platform)- (18.5 × 15 × 13 in) made from smooth, non-toxic acrylic with a ramp and UVB/heat lamp above. Safe, easy to clean, gives turtles more space and warmth, fits current 66.5-gallon tank. (mostly likely going to do this one)
2. Naturalistic Rock & Plant Habitat- Adds artificial rocks, live aquatic plants, and hiding spots to mimic a pond ecosystem while keeping a large basking area.
3. Smart Sensor System Heat and UVB sensors alert caretakers when levels drop. Ensures safety and consistent conditions.

D2 – Justification

We chose the Elevated Basking Platform v2.0 because it's the safest, easiest to clean, and most effective design for turtle health. It improves basking and UVB exposure, prevents injury, and fits the existing tank setup better than more complex options like the dual-zone tank.

D3 – Plan of Action

Step Task Who Materials Time

- 1 Sketch design & take measurements Hailey & Attia Paper, ruler 1 class period
- 2 Gather materials Jorge & team Acrylic, lamp, tools 1 day
- 3 Build & install platform Group Tools, glue 2 days
- 4 Test temp & UVB levels Group Thermometer, sensor 1 day
- 5 Observe turtles & adjust Group Notebook 3 days

D4 – Sketch & Details

Platform size: 18.5 × 15 × 13 in

Tank: 66.5 gallons

Lamp: 13W halogen + 5.0 UVB, 6 in above platform

Temp range: 90–95°F

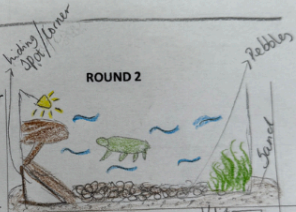
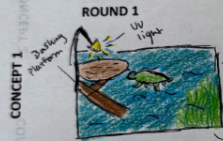
Sketch should show:

Ramp access, basking area under lamp, maybe adding another ramp (side note: my group still need to think about this)

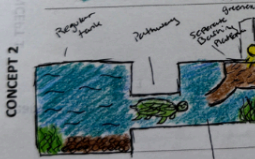
Group 8

Names Angelina Semand Serivera Lizzy

Team#: 8 Project:



ROUND 3



CONCEPT 3

Adding two ramps would be beneficial to the turtles well-being/fitness

Group 3

Names Pawans Ike, Angelina Semend

